Mathematics - Probability and Statistics

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Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Total Probability & Bayes' Theorem

Session 7

Total Probability & Bayes' Theorem

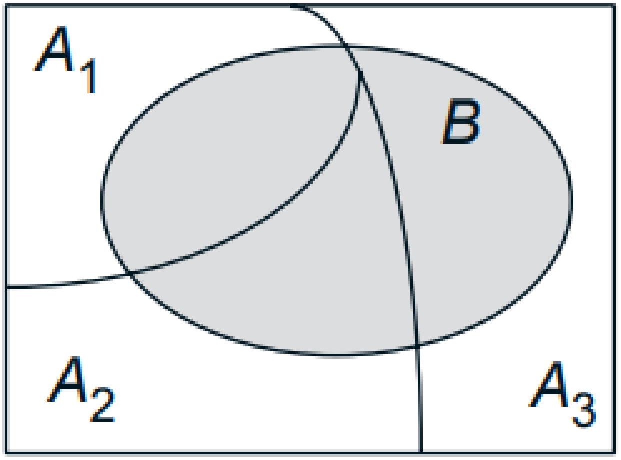
Session 7

Objectives

- Apply the Law of Total Probability to compute probabilities across partitions.
- Use probability trees to visualize conditional probabilities.
- Solve problems using Bayes' Theorem.
- Interpret results in diagnostic testing (medical tests, spam filters, reliability, etc.).

Law of Total Probability

The events $A_1,\ldots,A_n$ form a partition of the sample space. How can you obtain the probability of B?



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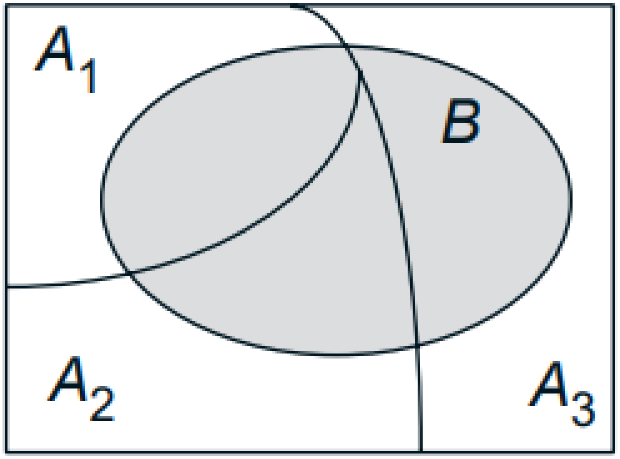


B can be decomposed into the disjoint union of its intersections $A_i \cap B$ with the sets A_i:

$$B = (A_1 \cap B) \cup \cdots \cup (A_n \cap B).$$

Law of Total Probability

$$B = (A_1 \cap B) \cup \cdots \cup (A_n \cap B).$$

Therefore, the proba is defined by:

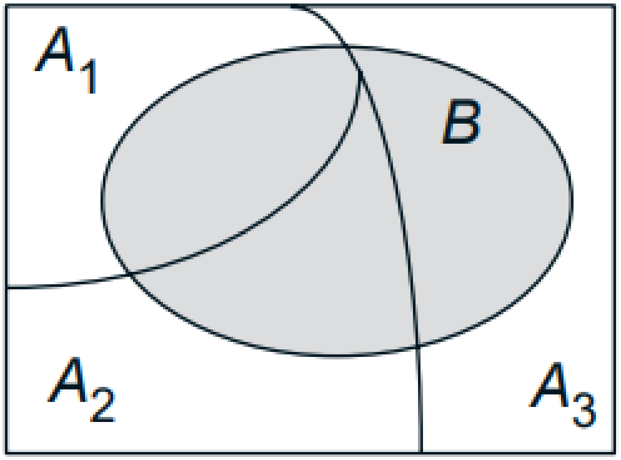
$$\mathbf{P}(B) = \mathbf{P}(A_1 \cap B) + \cdots + \mathbf{P}(A_n \cap B).$$

Because we have:

$$\mathbf{P}(A_i \cap B) = \mathbf{P}(A_i)\mathbf{P}(B \mid A_i),$$

Finally, we obtain:

$$\mathbf{P}(B) = \mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \cdots + \mathbf{P}(A_n)\mathbf{P}(B \mid A_n).$$



Law of Total Probability

Total Probability Theorem

Let $A_1, \ldots, A_n$ be disjoint events that form a partition of the sample space (each possible outcome is included in one and only one of the events $A_1, \ldots, A_n$) and assume that $\mathbf{P}(A_i) > 0$, for all $i = 1, \ldots, n$. Then, for any event B, we have

$$\begin{aligned}\mathbf{P}(B) &= \mathbf{P}(A_1 \cap B) + \cdots + \mathbf{P}(A_n \cap B)\\ &= \mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \cdots + \mathbf{P}(A_n)\mathbf{P}(B \mid A_n).\end{aligned}$$

Law of Total Probability

Exercices:

Three machines produce **metal rods**:

- Machine 1 produces **500 rods**.
- Machine 2 produces **800 rods**.
- Machine 3 produces **700 rods**.

The machines are imperfect and produce defective rods at different rates:

- Machine 1: **2% defective**.
- Machine 2: **4% defective**.
- Machine 3: **5% defective**.

Proba of taking a defective rod **at random among all 2000 rods**.

Law of Total Probability

Sol:

- Machine 1 produces **500 rods**. -> P(E1)=500/2000=0.25
- Machine 2 produces **800 rods**. -> P(E2)=800/2000=0.40
- Machine 3 produces **700 rods**. -> P(E3)=700/2000=0.35

If B is a event of selection a defective rod

P(B|E1)= 0.02
P(B|E2)= 0.04
P(B|E3)= 0.035

$$P(B) = \sum_{i=1}^{k} P(B \mid E_i)\, P(E_i).$$

P(B)=P(B|E1)*P(E1)+ P(B|E2)*P(E2)+ P(B|E3)*P(E3)
=0.0385

Tree decision

Steps

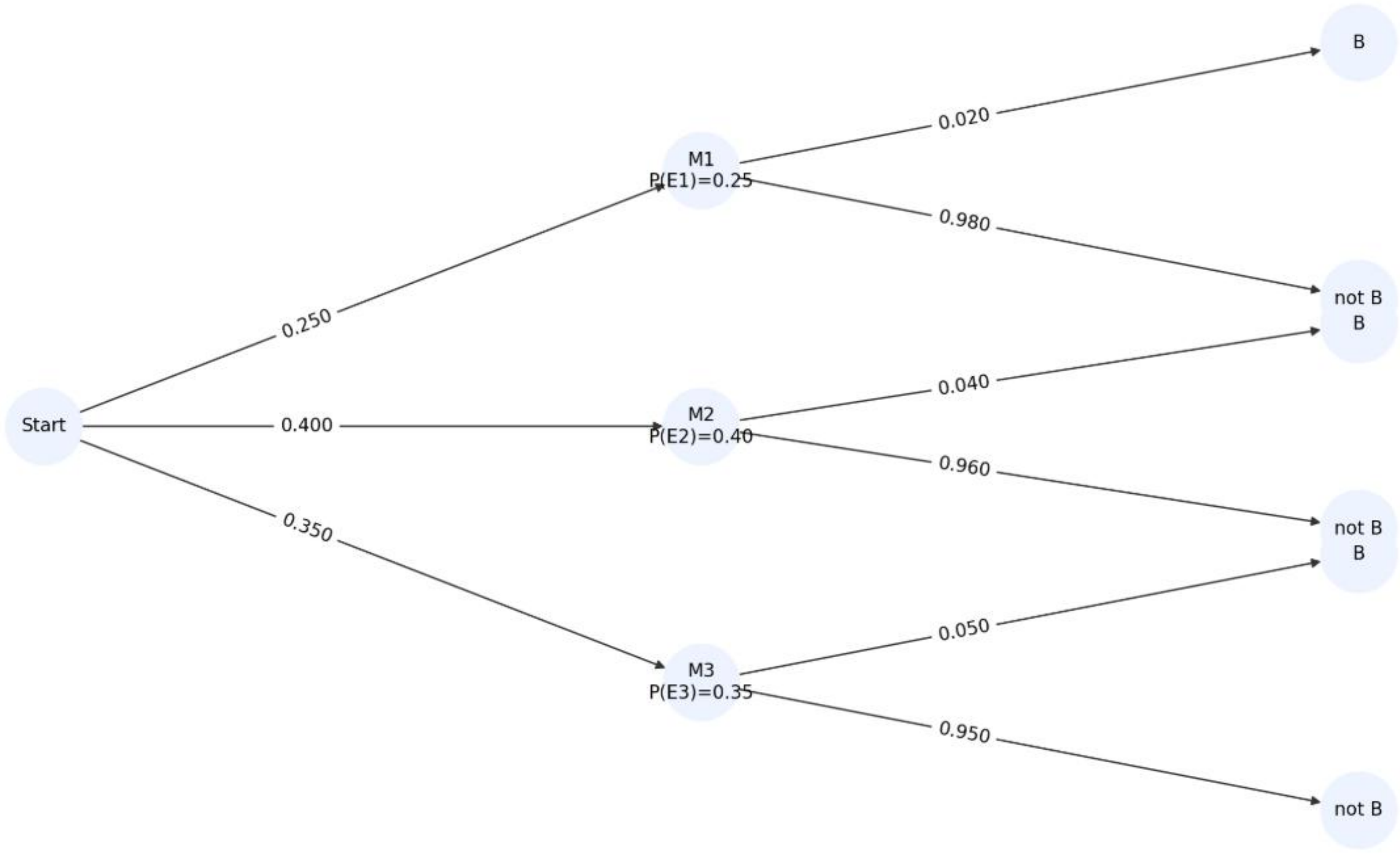
1.Root branches: B1,B2,...,Bk with P(Bi).

2.From each Bi, branch to A and Ac with P(A|Bi) and P(Ac|Bi)

3. Path probability = product of branch probabilities.

4.P(A)= **sum of path products** ending at A

Tree decision



0.250*0.02

0.40*0.04

0.35*0.05

0.0385

Law of Total Probability

Exercices:

You enter a chess tournament where your probability of winning a game

- 0.3 against half the players (red players)
- 0.4 against a quarter of the players (player blue),
- 0.5 against the remaining quarter of the players (yellow players).

You play a game against a randomly chosen opponent.

What is the probability of winning?

Law of Total Probability

Sol:

You enter a chess tournament where your probability of winning a game

- 0.3 against half the players (A1)
- 0.4 against a quarter of the players (A2),
- 0.5 against the remaining quarter of the players (A3).

You play a game against a randomly chosen opponent.

What is the probability of winning?

$$P(B) = \sum_{i=1}^{k} P(B \mid E_i)\, P(E_i).$$

If B is a event of winning

P(B|A1)= 0.2

P(B|A2)= 0.4

P(B|A3)= 0.5

P(A1)=0.5

P(A2)=P(A3)=0.25

P(B)=P(B|A1)*P(A1)+ P(B|A2)*P(A2)+ P(B|A3)*P(A3)

=0.375

Bayes Theorem

Problem:

A desk lamp was found to be defective. There are three factories:

Factory	% of total production	Probability of defective lamps
A	$0.35 = P(A)$	$0.015 = P(D\|A)$
B	$0.35 = P(B)$	$0.010 = P(D\|B)$
C	$0.30 = P(C)$	$0.020 = P(D\|C)$

If a randomly selected lamp is defective, what is the probability that the lamp was manufactured in factory C ?

Bayes Theorem

Solution:

If a randomly selected lamp is defective (D), what is the probability that the lamp was manufactured in factory C ?

Factory	% of total production	Probability of defective lamps
A	$0.35 = P(A)$	$0.015 = P(D\|A)$
B	$0.35 = P(B)$	$0.010 = P(D\|B)$
C	$0.30 = P(C)$	$0.020 = P(D\|C)$

P(C|D)= P(C n D) / P(D)

Using multiplication rule:

P(C|D)= P(D|C) * P(C) / P(D)

As P(D)= P(A n D) + P(B n D) + P(C n D)
And P(A nD) =P(A)*P(D|A) and the same for B and D we have finally:

P(C|D) = [P(D|C) * P(C)]/ [P(A)*P(D|A)+ P(B)*P(D|B)+ P(C)*P(D|C)]

=[0.02*0.3]/[(0.35*0.015+0.35*0.01+0.3*0.02]=0.407

Bayes Theorem

Bayes' theorem is very useful to solve P(B|A) in terms of P(A|B).

Bayes' Rule

- For any two events A and B, where $P(A) \neq 0$, we have

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}.$$

- If $B_1, B_2, B_3, \cdots$ form a partition of the sample space S, and A is any event with $P(A) \neq 0$, we have

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}.$$

The probabilities P(B) are often referred to as prior probabilities, because they are the probabilities of events , and that we know prior to obtaining any additional information. The conditional probabilities P(B|A), and are often referred to as posterior probabilities, because they are the probabilities of the events after we have obtained additional information.

Bayes Theorem

Exemple:

A certain disease affects about 1out of 10,000 people. There is a test to check whether the person has the disease. The test is quite accurate. In particular, we know that the probability that the test result is positive, given that the person does not have the disease, is only 2 percent; the probability that the test result is negative, given that the person has the disease, is only 1 percent.

A random person gets tested for the disease and the result comes back positive. What is the probability that the person has the disease?

Bayes Theorem

Solution:

D=event that the person has the disease
T= event that the test result is true/positive.

P(D)=1/10000
P(T|Dc)=0.02
P(Tc|D)=0.01 -> P(T|D)=0.99

We want P(D|T) ?

Bayes' Theorem:

P(D|T)=P(T|D)*P(D)/P(T)

P(T)=P(T|D)*P(D)+P(T|Dc)*P(Dc)=0.99*1/10000+0.02*9999/10000= 0.020097

So: P(D|T)=0.99*1/10000/ 0.020097= 0.0049=**0.5%**

$$P(B) = \sum_{i=1}^{k} P(B \mid E_i)\, P(E_i).$$

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}.$$

Exercice

Exercice:

Alice is taking a probability class and at the end of each week she can be either up-to-date or she may have fallen behind. If she is up-to-date in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.8 (or 0.2, respectively). If she is behind in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.6 (or 0.4, respectively). Alice is up-to-date when she starts the class. What is the probability that she is up-to-date after three weeks?

Exercice

Solution:

Alice is taking a probability class and at the end of each week she can be either up-to-date or she may have fallen behind. If she is up-to-date in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.8 (or 0.2, respectively). If she is behind in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.6 (or 0.4, respectively). Alice is up-to-date when she starts the class. What is the probability that she is up-to-date after three weeks?

Let $\mathbf{U_i}$ **and** $\mathbf{B_i}$ be the events that Alice is **up-to-date or behind**, respectively, after i weeks. According to the total probability theorem, the desired probability **P(U3)** is given by

$$P(B) = \sum_{i=1}^{k} P(B \mid E_i)\, P(E_i).$$

- $P(U3) = P(U2)P(U3 \mid U2) + P(B2)P(U3 \mid B2) = P(U2) \cdot 0.8 + P(B2) \cdot 0.6$

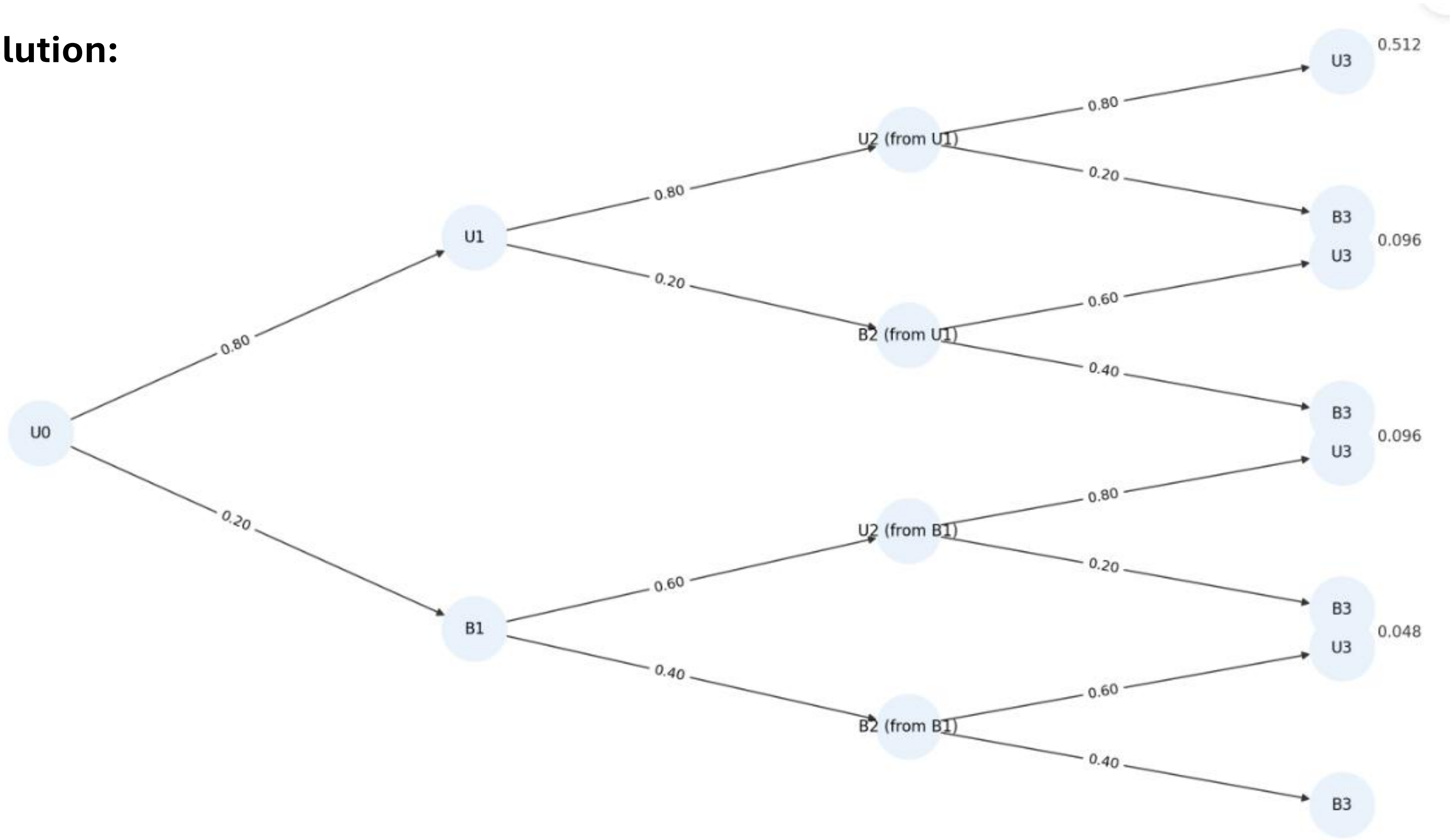
- $P(U2) = P(U1)P(U2 \mid U1) + P(B1)P(U2 \mid B1) = P(U1) \cdot 0.8 + P(B1) \cdot 0.6$
- $P(B2) = P(U1)P(B2 \mid U1) + P(B1)P(B2 \mid B1) = P(U1) \cdot 0.2 + P(B1) \cdot 0.4.$

$P(U2)=0.8 \cdot 0.8+0.2 \cdot 0.6=0.76$

$P(B2)=0.8 \cdot 0.2+0.2 \cdot 0.4=0.24$ therefore: $P(U3)=0.76 \cdot 0.8+0.24 \cdot 0.6=$ **0.752**

Exercice

Solution:



0.752

Exercice

Problem:

An insurance company believes that people can be divided into two classes — those that are accident prone and those that are not. Their statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a non-accident-prone person.

If we assume that 30 percent of the population is accident prone, what is the probability that a new policy holder will have an accident within a year of purchasing a policy?

Exercice

Solution: If we assume that 30 percent of the population is accident prone, what is the probability that a new policy holder will have an accident within a year of purchasing a policy?

D=Drivers prone to have accident
P(D)=0.3
A=having accident with a year

P(A|D)=0.4
P(A|Dc)=0.2

$$P(B) = \sum_{i=1}^{k} P(B \mid E_i)\, P(E_i).$$

P(A)=P(A|Dc)*P(Dc)+P(A|D)*P(D)=
=0.2*(1-0.3)+0.4*0.3
=**0.26**

Exercice

Problem:

The manual of a product states that the lifetime T, is defined as the amount of time (in years) the product works properly until it breaks down, and satisfies

$$\mathbf{P(T \geq t) = e^{-t/5}, \text{ for all } t \geq 0.}$$

I purchase the product and use it for two years without any problems. What is the probability that it breaks down in the third year?

Exercice

Solution: **$P(T \geq t) = e^{-t/5}$, for all $t \geq 0$.**

I purchase the product and use it for two years without any problems. What is the probability that it breaks down in the third year

A= event that a purchased product breaks down in the third year
B= event that a purchased product does not break down in the first two years

We want: $P(A|B)$

$P(A|B)=P(B|A)P(A)/P(B)$

$P(B)=e(-2/5)=0.6703$
$P(A)=P(2<T\leq 3)=e(-2/5)-e(-3/5)= 0.6703- 0.548=0.122$

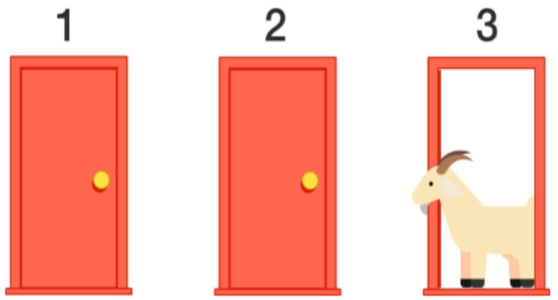
$P(B|A)$= proba that the event does break during the first 2 years if it breaks the third year……=1

$P(A|B)=1* 0.122/0.6703= 0.182=$ **18%**

Exercice

Problem: Monty Hall Problem

In the problem, you are on a game show, being asked to choose between **three doors**. Behind each door, there is either a car or a goat (2 goat 1 car). **You choose a door**. **The host, picks one of the other doors, which he knows has a goat behind it, and opens it, showing you the goat.** The host then asks whether you would like to switch your choice of door to the other remaining door.



Assuming you prefer having a car more than having a goat, do you choose to switch or not to switch?

What is the odd if you switch? What is the odd if you do not switch?

Exercice

Solution:

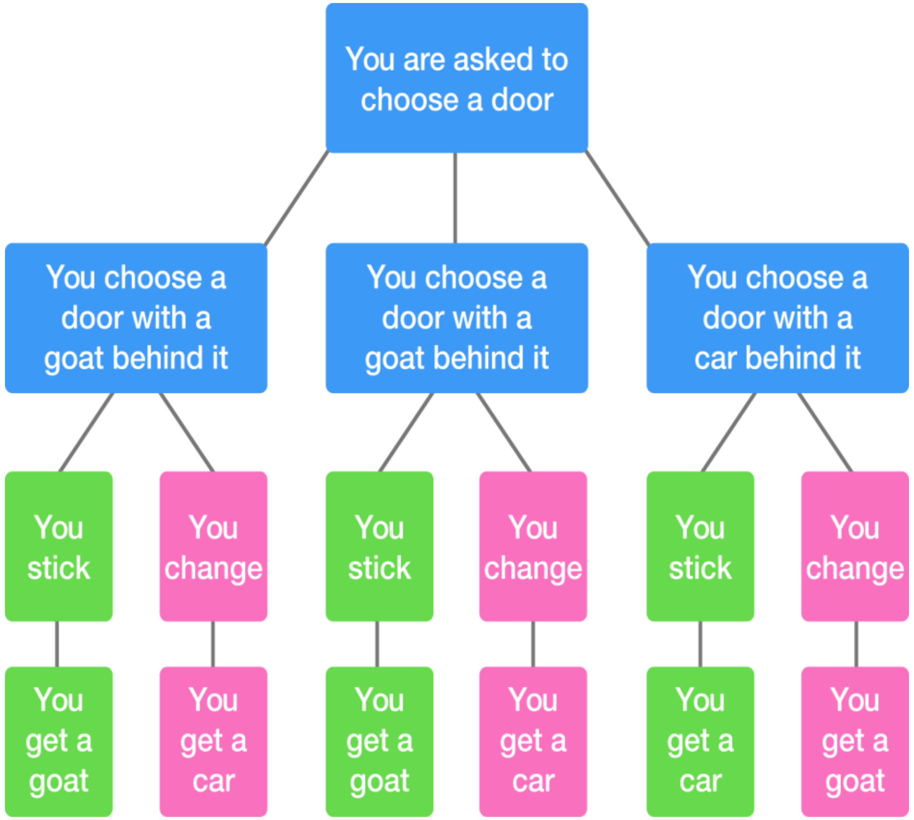
From the decision tree, we see that the probability
To have a car is 2/3 if you switch while only 1/3
If you stick.

Now let use Bayes theorem. With this theorem we can find if the probability that our initially picked door is the one with the car behind it given that the host has opened a door showing a goat behind it

Imagine you select door 1:

H =hypothesis "door 1 has a car behind it
E = evidence that host has revealed a door with a goat behind it

We want to find P(H|E) and will compare with the probability Is we change the door.



Exercice

Solution:

$$P(H \mid E) = \frac{P(E \mid H)}{P(E)} P(H) = \frac{P(E \mid H) \times P(H)}{P(E \mid H) \times P(H) + P(E \mid \text{not} H) \times P(\text{not} H)}.$$

- $P(H)$ is the prior probability that door 1 has a car behind it, without knowing about the door that Monty reveals. This is $\frac{1}{3}$.
- $P(\text{not} H)$ is the probability that we did not pick the door with the car behind it. Since the door either has the car behind it or not, $P(\text{not} H) = 1 - P(H) = \frac{2}{3}$.
- $P(E \mid H)$ is the probability that Monty shows a door with a goat behind it, given that there is a car behind door 1. Since Monty always shows a door with a goat, this is equal to 1.
- $P(E \mid \text{not} H)$ is the probability that Monty shows the goat, given that there is a goat behind door 1. Again, since Monty always shows a door with a goat, this is equal to 1.

$$P(H \mid E) = \frac{1 \times \frac{1}{3}}{1 \times \frac{1}{3} + 1 \times \frac{2}{3}} = \frac{\frac{1}{3}}{1} = \frac{1}{3}.$$

The probability that the car is behind door 1 is completely unchanged by the evidence. However, since the car can only either be behind door 1 or behind the door Monty didn't reveal, the probability it is behind the door that is not revealed is 1-1/3=**2/3**. Therefore, switching is twice as likely to get you the car as staying!!

Exercice

Problem:

Each bag in a large box contains 25 tulip bulbs. Three-fourths of the bags are of Type A containing bulbs for 5 red and 20 yellow tulips; one-fourth of the bags are of Type B contain bulbs for 15 red and 10 yellow tulips. A bag is selected at random and one bulb is planted. What is the probability that the bulb will produce a red tulip?

Exercice

Solution:

Each bag in a large box contains 25 tulip bulbs. Three-fourths of the bags are of Type A containing bulbs for 5 red and 20 yellow tulips; one-fourth of the bags are of Type B contain bulbs for 15 red and 10 yellow tulips. If the tulip is red, what is the probability that bag B was selected?

A=Type A. P(A)=3/4
B= Type B P(B)=1/4
R= tulip red

We want **P(B|R)**= P(R|B)P(B)/P(R)

- P(R|B)=15/25=3/5 and P(R|A)=1/5

P(R)= P[(AnR)U(BnR)]
Using multiplicative rule and mutually exclusive events

$$P(B) = \sum_i P(B \mid E_i)P(E_i).$$

P(R)=P(R|A)P(A)+P(R|B)P(B)=1/5*0.75+0.25*3/5=0.30

P(B|R)=1/5*1/4/(0.30)=**0.5**